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EDUCATION	Columbia University, New York, NY M.D./Ph.D., Medical Scientist Training Program (MSTP) University of Michigan, Ann Arbor, MI Master of Science in Mathematics Bachelor of Science in Mathematics	Aug 2026 – 2034 (Expected) Sep 2023 – Dec 2024 Sep 2020 – Dec 2024
RESEARCH EXPERIENCES	Columbia University, New York, NY Advisor: Dr. Orson Xu University of Michigan, Ann Arbor, MI Advisor: Dr. Daniel Forger Institute for Basic Science, Daejeon, South Korea Advisor: Dr. Jae Kyoung Kim	Jul 2026 – <i>present</i> Sep 2021 – Jul 2026 May 2022 – Aug 2022
RESEARCH INTERESTS	Computational Psychiatry, Digital Health, Human-Computer Interaction, Circadian Rhythms, Sleep	
PUBLICATIONS	* : co-first author, † : co-corresponding author Kim R*, Fang Y, Lee M , Kim DW, Tang Z, Sen S, Forger DB [†] , Seasonal timing and interindividual differences in shiftwork adaptation, <i>NPJ Digit. Med.</i> (2025). Mayer C*, Kim DW*, Zhang M, Lee M , Forger DB, Burgess HJ, Moon C [†] , Predicting circadian phase in community-dwelling later-life adults using actigraphy data, <i>J. Sleep Res.</i> (2024). Lee M *, Kim DW*, [†] Fang Y, Kim R, Bohnert ASB, Sen S, Forger DB [†] , The real-world association between digital markers of circadian disruption and mental health risks, <i>NPJ Digit. Med.</i> (2024). Lee M *, Kim DW*, [†] Mayer C, Walch O, Forger DB, The combination of topological data analysis and mathematical modeling improves sleep stage prediction from consumer-grade wearables, <i>J. Biol. Rhythms</i> (2024). Lee M *, Hoang K*, Park S, Song YM, Joo EY, Chang W [†] , Kim JH [†] , Kim JK [†] , Imputing missing sleep data from wearables with neural network in real-world settings, <i>Sleep</i> (2023). Kim DW*, Lee M *, Forger DB [†] , Wearable data assimilation to estimate the circadian phase, <i>SIAM J. Appl. Math</i> (2023). Kim DW*, [†] Mayer C*, Lee M , Choi SW, Tewari M, Forger DB, Efficient assessment of real-world dynamics of circadian rhythms in heart rate and body temperature from wearable data, <i>J. R. Soc. Interface</i> (2023). Lee M *, Kim R, Cochran A, Kim DW, Forger DB [†] , Mapping the landscape of behavior-induced circadian disruption across psychiatric disorders (2026+), Under review at <i>Nat. Hum. Behav.</i> Njoroge W*, Andai D, Lee M , Aballa A, Khakali L, Maina R, Frank E, Atwoli L, Wu Z, Ngugi AK, Sen S, Bunde A, Shah J, Mwigerari DG, Orwa J, Waljee AK, Abdukar A, Merali Z, Forger DB [†] , Feasibility of longitudinal sleep monitoring among Kenyan healthcare workers using wearable technology (2026+), Under review at <i>NPJ Biol. Timing Sleep</i> .	

HONORS AND AWARDS	2025 Student Travel Grant	National Institute for Theory and Mathematics in Biology
	2024 Best Poster Presentation Award	Society for Mathematical Biology
	2024 Rackham Graduate School International Travel Grant	University of Michigan
	2024-2025 Outstanding Achievement in Mathematics Award	University of Michigan
	2023 Barry M. Goldwater Scholarship	Goldwater Foundation
	2023 Wilfred Kaplan Award in Applied Mathematics	University of Michigan
	2020-2024 Department of Mathematics Merit Scholarship	University of Michigan
	2020 Andrew J. Lum & David R. Juillet Scholarship	Ann Arbor Community Foundation
PRESENTATION	2025 NSF-Simons Foundation NITMB MathBio Convergence Conference, Poster, Chicago, IL	
	2025 Biological Research Information Center Seminar, POSTECH, Online	
	2024 Society for Mathematical Biology Annual Meeting, Poster, Seoul, South Korea	
	2024 Society for Research on Biological Rhythms Biennial Meeting, Poster, San Juan, Puerto Rico	
	2023 SIAM Great Lakes Section Annual Meeting, Contributed talk, Lansing, MI, USA	
	2022 SIAM Great Lakes Section Annual Meeting, Minisymposium, Detroit, MI, USA	
	2022 IBS Biomedical Mathematics Seminar, Daejeon, South Korea	
	2022 Annual Conference of Korean Society for Industrial and Applied Mathematics, Poster, Daejeon, South Korea	
PEER REVIEW ACTIVITY	2021 University of Michigan Mathematics REU Seminar, Virtual	
	npj Digital Medicine, npj Biosensing, Critical Care, Scientific Reports	